Special Summer Integrated Learning Programme

Mathematics

Introduction to Integral Calculus

Week 6 Day 2 PM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

Partial fractions

Partial fractions is a technique used to integrate rational functions, i.e. $\int \frac{P(x)}{Q(x)}\,\mathrm{d}x$, where $P(x), Q(x)$ are polynomials. Here's the general procedure ($\deg(P)$ is the degree of the polynomial P):

1. If $\deg(P) \geq \deg(Q)$ the rational function is ***improper***. By long division, we can express it as a polynomial plus a ***proper*** rational function. For example, $\frac{x}{x^2 - x - 1}$ is proper and $\frac{3x^2 + x - 7}{x^2 - x - 1}$ is improper. So we can apply long division to express $\frac{3x^2 + x - 7}{x^2 - x - 1}$ as a sum of a polynomial and a proper rational function.

Partial fractions

1. By long division,

$$\begin{array}{r|l} & \phantom{3x^2 + x -{}} 3 \\ \hline x^2 - x - 1 & 3x^2 + x - 7 \\ & 3x^2 - 3x - 3 \\ \cline{2-2} & \phantom{3x^2 +{}} 4x - 4 \end{array}$$

   we have
   $$\frac{3x^2 + x - 7}{x^2 - x - 1} = 3 + \frac{4x - 4}{x^2 - x - 1}.$$
   So we can now find the partial fractions of the proper rational function $\dfrac{4x-4}{x^2-x-1}$.

2. $Q(x)$ can always be factorized in ***prime factors***, e.g. $x^4 - 2\,x^3 + 2\,x^2 - 2\,x + 1 = (x-1)^2\,(x^2+1)$, the same way natural numbers can be factorized in prime numbers, e.g. $1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7$.

Partial fractions

3. The prime factors are either polynomials of degree 1 or degree 2 without real roots.
4. We want to break $\frac{P(x)}{Q(x)}$ into simpler and smaller proper fractions which are easy to integrate.
5. **E.g.:**
$$\frac{1}{(x-2)\,(x^2+4)} = \frac{A}{x-2} + \frac{C\,x+D}{x^2+4}.$$
6. The RHS has three unknowns. For the RHS to equal to the LHS, we need three equations. They come from the equality of the numerators:

$$\frac{1}{(x-2)\,(x^2+4)} = \frac{A}{x-2} + \frac{Cx+D}{x^2+4} = \frac{A(x^2+4)+(Cx+D)(x-2)}{(x-2)(x^2+4)}.$$

$$0\,x^2 + 0\,x + 1 = (A+C)\,x^2 + (D-2\,C)\,x + 4\,A - 2\,D.$$

Partial fractions

7. We get a system of three equations with three unknowns:

$$\left.\begin{array}{ll} A + C & = 0 \\ D - 2\,C & = 0 \\ 4\,A - 2\,D & = 1. \end{array}\right\} \Rightarrow A = -C, D = 2C \Rightarrow -4C - 4C = 1 \Rightarrow C = -\frac{1}{8}.$$

8. So that

$$\frac{1}{(x-2)\left(x^2+4\right)} = \frac{1}{8}\,\frac{1}{x-2} - \frac{1}{8}\,\frac{x+2}{x^2+4} = \frac{1}{8}\,\frac{1}{x-2} - \frac{1}{8}\,\frac{x}{x^2+4} - \frac{1}{8}\,\frac{2}{x^2+4}.$$

9. And therefore

$$\begin{aligned} \int \frac{1}{(x-2)\left(x^2+4\right)}\,\mathrm{d}x &= \int \frac{1}{8}\,\frac{1}{x-2} - \frac{1}{8}\,\frac{x}{x^2+4} - \frac{1}{8}\,\frac{2}{x^2+4}\,\mathrm{d}x \\ &= \frac{1}{8}\,\ln|x-2| - \frac{1}{16}\,\ln\left(x^2+4\right) - \frac{1}{8}\,\arctan\frac{x}{2} + C. \end{aligned}$$

Partial fractions

For any $(x-a)^n$ and $(ax^2+bx+c)^n$ factors of $Q(x)$, you get the following terms:

$$\frac{P(x)}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \ldots + \frac{A_{n-1}}{(x-a)^{n-1}} + \frac{A_n}{(x-a)^n}$$

$$\frac{P(x)}{(ax^2+bx+c)^n} = \frac{B_1\,x+C_1}{ax^2+bx+c} + \frac{B_2\,x+C_2}{(ax^2+bx+c)^2} + \ldots + \frac{B_n\,x+C_n}{(ax^2+bx+c)^n}$$

For example:

$$\frac{x+1}{x^3(x^2+1)^2} = \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{A_3}{x^3} + \frac{B_1x+C_1}{x^2+1} + \frac{B_2x+C_2}{(x^2+1)^2}.$$

All these partial fractions can be integrated quite easily, hence we can compute $\int \frac{P(x)}{Q(x)}\mathrm{d}x$.

Partial fractions - example

E.g.: To compute $\int \frac{x^2+2}{x^2-7x+10}\,\mathrm{d}x$, we note that the integrand is an improper rational function. By long division, we have

$$\begin{array}{r|l} & 1 \\ \cline{2-2} x^2-7x+10 & x^2+0x+\ \ 2 \\ & \underline{x^2-7x+10} \\ & 7x-\ \ 8 \end{array}$$

we have

$$\frac{x^2+2}{x^2-7x+10} = 1 + \frac{7x-8}{x^2-7x+10}.$$

The denominator can be factorized as
$x^2-7x+10=(x-2)(x-5)$, so

$$\int \frac{x^2+2}{x^2-7x+10}\,\mathrm{d}x = \int \left(1+\frac{7x-8}{(x-2)(x-5)}\right)\mathrm{d}x$$

Partial fractions - example

We apply partial fractions to the proper rational function $\dfrac{7x-8}{x^2-7x+10} = \dfrac{7x-8}{(x-2)(x-5)}$ and rewrite

$$\begin{aligned}\frac{7x-8}{(x-2)(x-5)} &= \frac{A}{x-2} + \frac{B}{x-5} \\ &= \frac{Ax-5A+Bx-2B}{(x-2)(x-5)} \\ &= \frac{(A+B)x-5A-2B}{(x-2)(x-5)}.\end{aligned}$$

So $A+B=7$ and $-5A-2B=-8$. They imply $A=7-B$ and so $-5(7-B)-2B=-8 \;\Rightarrow\; 3B=27 \;\Rightarrow\; B=9$ and $A=7-9=-2$. Hence,

$$\frac{7x-8}{(x-2)(x-5)} = \frac{-2}{x-2} + \frac{9}{x-5}.$$

Partial fractions - example

Finally, we can compute the integral:

$$\begin{aligned}\int \frac{x^2+2}{x^2-7x+10}\,\mathrm{d}x &= \int \left(1+\frac{7x-8}{(x-2)(x-5)}\right)\mathrm{d}x \\ &= \int \left(1+\frac{-2}{x-2}+\frac{9}{x-5}\right)\mathrm{d}x \\ &= x-2\ln|x-2|+9\ln|x-5|+C.\end{aligned}$$

Partial fractions - example

E.g.: To compute $\int \frac{x^3+2x^2+x+1}{x(x^2+1)^2}\,\mathrm{d}x$, we note that the integrand is a proper rational function and the denominator is factorized in prime factors, so we proceed to apply partial fractions and rewrite

$$\frac{x^3+2x^2+x+1}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}. \tag{1}$$

Multiplying equation (1) by x gives

$$\frac{x^3+2x^2+x+1}{(x^2+1)^2} = A + \frac{(Bx+C)x}{x^2+1} + \frac{(Dx+E)x}{(x^2+1)^2}.$$

By substituting $x=0$, we get

$$\frac{0+1}{(0+1)^2} = A+0+0 \;\Leftrightarrow\; A=1.$$

Partial fractions - example

$$\frac{0x^4 + 1\cdot x^3 + 2x^2 + 1\cdot x + 1}{x(x^2+1)^2}$$
$$= \frac{1}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$
$$= \frac{(x^2+1)^2 + (Bx+C)(x^2+1)x + (Dx+E)x}{x(x^2+1)^2}$$
$$= \frac{x^4 + 2x^2 + 1 + Bx^4 + Bx^2 + Cx^3 + Cx + Dx^2 + Ex}{x(x^2+1)^2}$$
$$= \frac{(1+B)x^4 + Cx^3 + (2+B+D)x^2 + (C+E)x + 1}{x(x^2+1)^2}.$$

So we have $1+B=0 \;\Rightarrow\; B=-1$, $C=1$,
$2+B+D=2 \;\Rightarrow\; D=2-2-B=2-2+1=1$ and
$C+E=1 \;\Rightarrow\; E=1-C=0$. So

$$\frac{x^3+2x^2+x+1}{x(x^2+1)^2} = \frac{1}{x} + \frac{1-x}{x^2+1} + \frac{x}{(x^2+1)^2}$$

Partial fractions - example

Finally, we can compute the integral:

$$\int \frac{x^3+2x^2+x+1}{x(x^2+1)^2}\,\mathrm{d}x = \int \left(\frac{1}{x} + \frac{1-x}{x^2+1} + \frac{x}{(x^2+1)^2}\right)\mathrm{d}x$$
$$= \int \frac{1}{x}\,\mathrm{d}x + \int \frac{1}{x^2+1}\,\mathrm{d}x - \int \frac{x}{x^2+1}\,\mathrm{d}x + \int \frac{x}{(x^2+1)^2}\,\mathrm{d}x$$
$$= \ln|x| + \arctan x - \int \frac{x}{x^2+1}\,\mathrm{d}x + \int \frac{x}{(x^2+1)^2}\,\mathrm{d}x.$$

The last two integrals can be solved by substituting $u = x^2+1 \;\Rightarrow\; \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \;\Rightarrow\; x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$. So

$$\int \frac{x^3+2x^2+x+1}{x(x^2+1)^2}\,\mathrm{d}x = \ln|x| + \arctan x - \int \frac{1}{u}\cdot\frac{1}{2}\,\mathrm{d}u + \int \frac{1}{u^2}\cdot\frac{1}{2}\,\mathrm{d}u$$
$$= \ln|x| + \arctan x - \frac{1}{2}\ln|u| - \frac{1}{2}\cdot\frac{1}{u} + C$$
$$= \ln|x| + \arctan x - \frac{1}{2}\ln(x^2+1) - \frac{1}{2(x^2+1)} + C$$

Activity 1

Use partial fractions to find the following integrals.

(a) $\displaystyle\int \frac{x^2+2}{x^2+x}\,\mathrm{d}x.$

(b) $\displaystyle\int \frac{4x^3-8x^2+3x-1}{x^2(x-1)^2}\,\mathrm{d}x.$

(c) $\displaystyle\int \frac{2}{x^4-1}\,\mathrm{d}x.$

(d) $\displaystyle\int \frac{3}{x^4+13\,x^2+36}\,\mathrm{d}x.$

Activity 1: solution

(a) We note that the integrand is an improper rational function. By long division, we have

$$\begin{array}{r|l} & \;1 \\ \hline x^2 + x + 0 & x^2 + 0x + 2 \\ & x^2 + x + 0 \\ \hline & -x + 2 \end{array}$$

we have

$$\frac{x^2 + 2}{x^2 + x} = 1 + \frac{2 - x}{x^2 + x}.$$

The denominator can be factorized as $x^2 + x = x(x + 1)$, so

$$\int \frac{x^2 + 2}{x^2 + x} \, \mathrm{d}x = \int \left(1 + \frac{2 - x}{x(x + 1)} \right) \mathrm{d}x$$

Activity 1: solution

(a) We apply partial fractions to the proper rational function $\dfrac{2-x}{x(x+1)}$ and rewrite

$$\begin{aligned}\frac{2-x}{x(x+1)} &= \frac{A}{x}+\frac{B}{x+1}\\ &= \frac{Ax+A+Bx}{x(x+1)}\\ &= \frac{(A+B)x+A}{x(x+1)}.\end{aligned}$$

So $A+B=-1$ and $A=2$. They imply $B=-1-A=-3$. Hence,

$$\frac{2-x}{x(x+1)}=\frac{2}{x}-\frac{3}{x+1}.$$

Activity 1: solution

(a) Finally, we can compute the integral:

$$\begin{aligned}\int \frac{x^2+2}{x^2+x}\,\mathrm{d}x &= \int \left(1+\frac{2-x}{x(x+1)}\right)\mathrm{d}x\\ &= \int \left(1+\frac{2}{x}-\frac{3}{x+1}\right)\mathrm{d}x\\ &= x+2\ln|x|-3\ln|x+1|+C.\end{aligned}$$

(b) Since $\dfrac{4x^3-8x^2+3x-1}{x^2(x-1)^2}$ is proper, by partial fractions, we have

$$\begin{aligned}\frac{4x^3-8x^2+3x-1}{x^2(x-1)^2} &= \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}+\frac{D}{(x-1)^2}\\ &= \frac{Ax(x-1)^2+B(x-1)^2+Cx^2(x-1)+Dx^2}{x^2(x-1)^2}.\end{aligned}$$

Activity 1: solution

(b) So we have the equality of numerators

$$4x^3-8x^2+3x-1 = Ax(x-1)^2+B(x-1)^2+Cx^2(x-1)+Dx^2.$$

Substituting $x = 1$ to the equation above gives
$4-8+3-1 = 0+0+0+D \ \Rightarrow \ D = -2$, whereas
substituting $x = 0$ gives
$0-0+0-1 = 0+B(0-1)^2+0+0 \ \Rightarrow \ B = -1$. So we have

$$\begin{aligned}
&4x^3-8x^2+3x-1 \\
&= Ax(x-1)^2-(x-1)^2+Cx^2(x-1)-2x^2 \\
&= Ax^3-2Ax^2+Ax-(x^2-2x+1)+Cx^3-Cx^2-2x^2 \\
&= (A+C)x^3-(2A+3+C)x^2+(A+2)x-1.
\end{aligned}$$

So we have $A+C=4$, $2A+3+C=8$ and
$A+2=3 \ \Rightarrow \ A=1$, which gives $C=4-A=4-1=3$.

Activity 1: solution

(b) Thus, we have

$$\int \frac{4x^3 - 8x^2 + 3x - 1}{x^2(x-1)^2}\,\mathrm{d}x = \int \frac{1}{x} - \frac{1}{x^2} + \frac{3}{x-1} - \frac{2}{(x-1)^2}\,\mathrm{d}x$$
$$= \ln|x| + \frac{1}{x} + 3\ln|x-1| + \frac{2}{x-1} + C.$$

(c) Since the proper rational function

$$\frac{2}{x^4-1} = \frac{2}{(x^2-1)(x^2+1)} = \frac{2}{(x-1)(x+1)(x^2+1)},$$ so

$$\frac{2}{x^4-1} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$
$$= \frac{A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x^2-1)}{x^4-1}.$$

So we have the equality of numerators

$$2 = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x^2-1).$$

Substituting $x = 1$ gives $2 = A(2)(2) + 0 + 0 \;\Rightarrow\; A = \frac{1}{2}$.

Activity 1: solution

(c) $2 = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x^2-1)$.

Substituting $x = -1$ gives
$2 = 0 + B(-2)(2) + 0 \;\Rightarrow\; B = -\frac{1}{2}$. So

$$\begin{aligned}
\frac{2}{x^4-1} &= \frac{1}{2}\cdot\frac{1}{x-1} - \frac{1}{2}\cdot\frac{1}{x+1} + \frac{Cx+D}{x^2+1} \\
&= \frac{1}{2}\cdot\frac{x+1-(x-1)}{x^2-1} + \frac{Cx+D}{x^2+1} \\
&= \frac{1}{x^2-1} + \frac{Cx+D}{x^2+1} \\
&= \frac{x^2+1+(Cx+D)(x^2-1)}{x^4-1} \\
&= \frac{x^2+1+Cx^3+Dx^2-Cx-D}{x^4-1} \\
&= \frac{Cx^3+(D+1)x^2-Cx+1-D}{x^4-1},
\end{aligned}$$

which gives $C = 0$ and $D = -1$.

Activity 1: solution

(c) Then

$$\int \frac{2}{x^4-1}\,\mathrm{d}x = \int \frac{1}{2}\cdot\frac{1}{x-1} - \frac{1}{2}\cdot\frac{1}{x+1} - \frac{1}{x^2+1}\,\mathrm{d}x$$
$$= \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| - \arctan x + C.$$

(d) We can substitute $y = x^2$ and hence

$$\frac{3}{x^4+13\,x^2+36} = \frac{3}{y^2+13y+36} = \frac{3}{(y+4)(y+9)} = \frac{A}{y+4}+\frac{B}{y+9},$$

which has only two unknowns A, B. This works because $x^4+13\,x^2+36$ doesn't have odd powers of x. Since

$$\frac{3}{(y+4)(y+9)} = \frac{A(y+9)+B(y+4)}{(y+4)(y+9)} = \frac{(A+B)y+9A+4B}{(y+4)(y+9)},$$

it implies $A+B=0$ and $9A+4B=3$,
$\Rightarrow\ 5A+4(A+B)=3\ \Rightarrow\ 5A+0=3 \Rightarrow\ A=\frac{3}{5}$,
$B=-A=-\frac{3}{5}$.

Activity 1: solution

(d) So

$$\int \frac{3}{x^4+13\,x^2+36}\mathrm{d}x = \frac{3}{5}\int \frac{1}{x^2+4}\mathrm{d}x - \frac{3}{5}\int \frac{1}{x^2+9}\mathrm{d}x.$$

We apply the result $\displaystyle\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\frac{x}{a} + C$ from activity 3(g) of the previous class and get

$$\begin{aligned}
\int \frac{3}{x^4+13\,x^2+36}\mathrm{d}x &= \frac{3}{5}\int \frac{1}{x^2+4}\mathrm{d}x - \frac{3}{5}\int \frac{1}{x^2+9}\mathrm{d}x \\
&= \frac{3}{5}\int \frac{1}{x^2+2^2}\mathrm{d}x - \frac{3}{5}\int \frac{1}{x^2+3^2}\mathrm{d}x \\
&= \frac{3}{5}\cdot\frac{1}{2}\arctan\frac{x}{2} - \frac{3}{5}\cdot\frac{1}{3}\arctan\frac{x}{3} + C \\
&= \frac{3}{10}\arctan\frac{x}{2} - \frac{1}{5}\arctan\frac{x}{3} + C.
\end{aligned}$$

Outline

1. Integration by Partial Fractions
2. Integration by Parts
3. Summary

Integration by parts

Integration by parts is a powerful integration technique for solving integrations of which the integrand is a product.

Since the derivative of a product is not the product of the derivatives, then

$$\int f(x)\,h(x)\,\mathrm{d}x \neq \int f(x)\,\mathrm{d}x \int h(x)\,\mathrm{d}x + C.$$

Recall the product rule:

$$\begin{aligned} & \frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\,g(x)\right) = f(x)\,g'(x) + f'(x)\,g(x) \\ \Rightarrow \quad & f(x)\,g'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\,g(x)\right) - f'(x)\,g(x). \end{aligned}$$

Integrating the equation above gives the formula of integration by parts:

$$\int f(x)\,g'(x)\,\mathrm{d}x = f(x)\,g(x) - \int f'(x)\,g(x)\,\mathrm{d}x.$$

Integration by parts

$$\int f(x)\, g'(x)\, \mathrm{d}x = f(x)\, g(x) - \int f'(x)\, g(x)\, \mathrm{d}x. \qquad (2)$$

- Let $g'(x) = h(x)$ and so $g(x) = \int h(x)\, \mathrm{d}x$. (2) becomes

$$\int f(x)\, h(x)\, \mathrm{d}x = f(x) \int h(x)\, \mathrm{d}x - \int f'(x) \left(\int h(x)\, \mathrm{d}x \right) \mathrm{d}x.$$

- If we let $u = f(x), \mathrm{d}u = f'(x)\, \mathrm{d}x$ and $v = g(x)$, $\mathrm{d}v = g'(x)\, \mathrm{d}x$, then (2) becomes

$$\int u\, \mathrm{d}v = u\, v - \int v\, \mathrm{d}u.$$

Both formulas mean the same thing. Choose the one that you can remember.

Guidelines for integration by parts

$$\int f(x)\,h(x)\,\mathrm{d}x = f(x)\int h(x)\,\mathrm{d}x - \int f'(x)\left(\int h(x)\,\mathrm{d}x\right)\mathrm{d}x.$$

- Notice that there's still an integration on the right hand side. So when we apply integration by parts, the aim is to change the original integration (the one on the left hand side) to a simpler integration (the one on the right hand side).
- The formula requires you to differentiate f and integrate h, so choose the $h(x)$ that you can integrate.
- If you can integrate both functions in the product, then choose the $f(x)$ such that $f'(x)$ is less complicated.

Integration by parts - example

$$\int f(x)h(x)\,\mathrm{d}x = f(x)\int h(x)\,\mathrm{d}x - \int f'(x)\left(\int h(x)\,\mathrm{d}x\right)\,\mathrm{d}x.$$

E.g.: $\int x \cdot \sin(x)\,\mathrm{d}x$.

We have x times $\sin(x)$, which one should we differentiate and which one should we integrate?

Let $f(x) = x$ and $h(x) = \sin(x)$, then $\int h(x)\,\mathrm{d}x = -\cos(x) + c_1$.

So

$$\begin{aligned}\int x \cdot \sin(x)\,\mathrm{d}x &= x(-\cos(x) + c_1) - \int (-\cos(x) + c_1)\,\mathrm{d}x \\ &= -x \cdot \cos(x) + x \cdot c_1 + \sin(x) - x \cdot c_1 + c_2 \\ &= -x \cdot \cos(x) + \sin(x) + c_2\end{aligned}$$

Note that any c_1 will work as we just need any antiderivative of $h(x)$. So just choose $c_1 = 0$ for convenience.

Integration by parts - example

$$\int f(x)h(x)\,\mathrm{d}x = f(x)\int h(x)\,\mathrm{d}x - \int f'(x)\left(\int h(x)\,\mathrm{d}x\right)\mathrm{d}x.$$

A short way of writing the previous integral is the following:

$$\int x{\cdot}\sin(x)\,\mathrm{d}x = x(-\cos(x)) - \int 1{\cdot}(-\cos(x))\,\mathrm{d}x = -x\cos(x) + \sin(x) + c.$$

Could we have used $f(x) = \sin(x)$ and $h(x) = x$? Yes, but it would not simplify the integral!

$$\int x \cdot \sin(x)\,\mathrm{d}x = \sin(x)\frac{x^2}{2} - \int \cos(x)\frac{x^2}{2}\,\mathrm{d}x.$$

This does not help! So it is important to pick the right functions!

Integration by parts - example

$$\int u\,\mathrm{d}v = u\,v - \int v\,\mathrm{d}u.$$

E.g.: $\int x^2\,e^x\,\mathrm{d}x.$

Suppose we choose $x^2\,\mathrm{d}x = \mathrm{d}v$. Then $v = \frac{x^3}{3}$, $u = e^x$ and $\mathrm{d}u = e^x\,\mathrm{d}x$. Integration by parts gives

$$\int x^2\,e^x\,\mathrm{d}x = \frac{x^3}{3}\,e^x - \frac{1}{3}\int x^3\,e^x\,\mathrm{d}x.$$

The second integral is more difficult than the original one. This is not helpful.

Let's choose $e^x\,\mathrm{d}x = \mathrm{d}v$. Then $v = e^x$, $x^2 = u$ and $\mathrm{d}u = 2\,x\,\mathrm{d}x$. Integration by parts gives

$$\int x^2\,e^x\,\mathrm{d}x = x^2\,e^x - 2\int x\,e^x\,\mathrm{d}x.$$

The second integral is simpler than the original one.

Integration by parts - example

$$\int u\,\mathrm{d}v = u\,v - \int v\,\mathrm{d}u.$$

Now we want to find $\int x\,e^x\,\mathrm{d}x$.

Choose $e^x\,\mathrm{d}x = \mathrm{d}v$. Then $v = e^x$, $x = u$ and $\mathrm{d}u = \mathrm{d}x$.

$$\int x\,e^x\,\mathrm{d}x = x\,e^x - \int e^x\,\mathrm{d}x = x\,e^x - e^x + c.$$

Finally

$$\int x^2\,e^x\,\mathrm{d}x = x^2\,e^x - 2\,(x\,e^x - e^x) + C.$$

Remark: If we chose $\mathrm{d}v = x\,\mathrm{d}x$ and $u = e^x$ when we applied integration by parts the second time, we would get back the original integral and made zero progress!

Integration by parts - example

E.g.: $\int \sin^3 x \,\mathrm{d}x.$

Choose $\sin x \,\mathrm{d}x = \mathrm{d}v$. Then $v = -\cos x$, $u = \sin^2 x$ and $\mathrm{d}u = 2 \sin x \ \cos x \,\mathrm{d}x$.

$$\begin{aligned}\int \sin^3 x \,\mathrm{d}x &= \int \sin^2 x \ \sin x \,\mathrm{d}x \\ &= -\cos x \ \sin^2 x + 2 \int \cos^2 x \ \sin x \,\mathrm{d}x, \\ &= -\cos x \ \sin^2 x - \frac{2}{3} \cos^3 x + C,\end{aligned}$$

where we use the substitution $t = \cos x$ to compute the last integral.

Guidelines for integration by parts

Generally speaking the functions lower on the following list have easier antiderivatives. So usually we'll choose the function lower on the list to be $\mathrm{d}v$ and the other function higher on the list to be u.

1. Logarithmic functions ($\ln(x)$).
2. Inverse trigonometric functions ($\arcsin(x)$, etc.).
3. Algebraic functions (x^n, polynomials).
4. Trigonometric functions ($\cos(x)$, etc.).
5. Exponential functions (e^x, a^x).

Guidelines for integration by parts

$$\int f(x)h(x)\,\mathrm{d}x = f(x)\int h(x)\,\mathrm{d}x - \int f'(x)\left(\int h(x)\,\mathrm{d}x\right)\mathrm{d}x.$$

Sometimes the integrand is not a product and you might want to use $h(x) = 1$.

E.g.: $\displaystyle\int \ln(x)\,\mathrm{d}x$.

We can think of $\ln(x) = \ln(x)\cdot 1$ and choose and $f(x) = \ln(x)$, $h(x) = 1$, so $f'(x) = \frac{1}{x}$ and $\displaystyle\int h(x)\,\mathrm{d}x = x$.

$$\begin{aligned}\int \ln(x)\,\mathrm{d}x &= \int \ln(x)\cdot 1\,\mathrm{d}x \\ &= \ln(x)\cdot x - \int \frac{1}{x}\cdot x\,\mathrm{d}x \\ &= \ln(x)\cdot x - x + c.\end{aligned}$$

ACTIVITY 2

Use integration by parts to find the following integrals.

(a) $\int \arcsin x \, \mathrm{d}x.$

(b) $\int x^6 \ln x \, \mathrm{d}x.$

(c) $\int x^2 \sin 4x \, \mathrm{d}x.$

(d) $\int \frac{x}{e^x} \, \mathrm{d}x.$

Activity 2: solution

(a) We can think of $\arcsin x = \arcsin x \cdot 1$ and choose $f(x) = \arcsin x$, $h(x) = 1$, so $f'(x) = \frac{1}{\sqrt{1-x^2}}$ and $\int h(x)\,\mathrm{d}x = x$.

$$\int \arcsin x \,\mathrm{d}x = \int \arcsin x \cdot 1 \,\mathrm{d}x$$
$$= \arcsin x \cdot x - \int \frac{1}{\sqrt{1-x^2}} \cdot x \,\mathrm{d}x.$$

For the second integration, we substitute $y = 1 - x^2 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -2x \;\Rightarrow\; x\,\mathrm{d}x = -\frac{1}{2}\,\mathrm{d}y$, so

$$\begin{aligned}\int \arcsin x \,\mathrm{d}x &= \arcsin x \cdot x - \int \frac{1}{\sqrt{1-x^2}} \cdot x \,\mathrm{d}x \\ &= x \arcsin x - \int \frac{1}{\sqrt{y}} \cdot \left(-\frac{1}{2}\right) \mathrm{d}y \\ &= x \arcsin x + \frac{1}{2}\frac{1}{\frac{1}{2}}\sqrt{y} + C = x \arcsin x + \sqrt{1-x^2} + C.\end{aligned}$$

Activity 2: solution

(b) Following the guidelines, we choose $u = \ln x \;\Rightarrow\; \mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$ and $\mathrm{d}v = x^6 \;\Rightarrow\; v = \frac{x^7}{7}$. By integration by parts, we get

$$\begin{aligned}\int x^6 \ln x \,\mathrm{d}x &= \frac{x^7}{7}\ln x - \int \frac{x^7}{7}\cdot\frac{1}{x}\,\mathrm{d}x \\ &= \frac{x^7}{7}\ln x - \int \frac{x^6}{7}\,\mathrm{d}x \\ &= \frac{x^7}{7}\ln x - \frac{x^7}{49} + C.\end{aligned}$$

Activity 2: solution

(c) Following the guidelines, we choose $u = x^2 \;\Rightarrow\; \mathrm{d}u = 2x\,\mathrm{d}x$ and $\mathrm{d}v = \sin 4x\,\mathrm{d}x \;\Rightarrow\; v = -\frac{1}{4}\cos 4x$. By integration by parts, we get

$$\begin{aligned}\int x^2 \sin 4x\,\mathrm{d}x &= -\frac{1}{4}\cos 4x \cdot x^2 + \int \frac{1}{4}\cos 4x \cdot (2x)\,\mathrm{d}x \\ &= -\frac{x^2}{4}\cos 4x + \frac{1}{2}\int x\cos 4x\,\mathrm{d}x.\end{aligned}$$

We apply integration by parts again on the second integral, with $u = x$ and $\mathrm{d}v = \cos 4x\,\mathrm{d}x$. Hence,

$$\begin{aligned}\int x^2 \sin 4x\,\mathrm{d}x &= -\frac{x^2}{4}\cos 4x + \frac{1}{2}\int x\cos 4x\,\mathrm{d}x \\ &= -\frac{x^2}{4}\cos 4x + \frac{1}{2}\left(x\frac{1}{4}\sin 4x - \int \frac{1}{4}\sin 4x\,\mathrm{d}x\right) \\ &= -\frac{x^2}{4}\cos 4x + \frac{x}{8}\sin 4x + \frac{1}{32}\cos 4x + C.\end{aligned}$$

Activity 2: solution

(d) Since $\frac{x}{e^x} = xe^{-x}$, we follow the guidelines and choose $u = x \;\Rightarrow\; \mathrm{d}u = \mathrm{d}x$ and $\mathrm{d}v = e^{-x}\,\mathrm{d}x \;\Rightarrow\; v = -e^{-x}$. By integration by parts, we get

$$\begin{aligned}\int \frac{x}{e^x}\,\mathrm{d}x &= \int xe^{-x}\,\mathrm{d}x \\ &= -xe^{-x} - \int 1\cdot(-e^{-x})\,\mathrm{d}x \\ &= -xe^{-x} - e^{-x} + C.\end{aligned}$$

Integration by parts for definite integrals

$$\int_a^b f(x)h(x)\,\mathrm{d}x = \left(f(x)\int h(x)\,\mathrm{d}x\right)\Bigg|_a^b - \int_a^b f'(x)\left(\int h(x)\,\mathrm{d}x\right)\mathrm{d}x$$

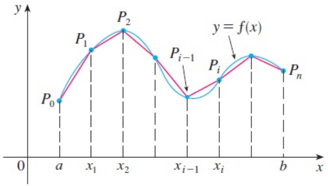
E.g.:

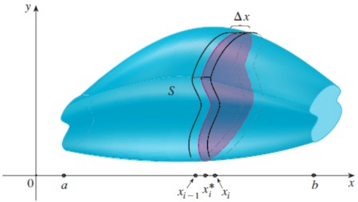
$$\begin{aligned}\int_1^e x^8 \ln x\,\mathrm{d}x &= \frac{x^9}{9}\ln x\Bigg|_1^e - \int_1^e \frac{x^9}{9}\frac{1}{x}\,\mathrm{d}x \\ &= \left(\frac{e^9}{9}\ln e - \frac{1^9}{9}\ln 1\right) - \frac{1}{9}\int_1^e x^8\,\mathrm{d}x \\ &= \frac{e^9}{9} - \frac{x^9}{9^2}\Bigg|_1^e = \frac{e^9}{9} - \frac{e^9}{9^2} + \frac{1}{9^2} = \frac{8e^9+1}{81}.\end{aligned}$$

Applications of integration

Besides having the geometrical meaning as the signed area between the graph and the x-axis, the general application of definite integrals is to compute sums of small changes.

Examples of sums of small changes in geometry are the volume of an object with varying cross section, the arc length of a curve and the area of a surface of revolution. Other examples of sums of small changes in physics are displacements, velocities, the work done by a varying force, the mass of an object with varying density, the center of mass, the moment of inertia, etc.





Last words

"Without calculus, we wouldn't have cell phones, computers, or microwave ovens. We wouldn't have radio. Or television. Or ultrasound for expectant mothers, or GPS for lost travelers. We wouldn't have split the atom, unraveled the human genome, or put astronauts on the moon. We might not even have the Declaration of Independence."

Steven Strogatz, *Infinite Powers: How Calculus Reveals the Secrets of the Universe*

Outline

Summary

- Integration by Partial Fractions
  - improper and proper rational functions, long division.
  - prime factors of polynomials.
  - partial fractions.
- Integration by Parts
  - the formula of integraion by parts.
  - guidelines for integration by parts - choosing the right f and h, or u and v.
  - LIATE, rewriting $f(x) = f(x) \cdot 1$.

Resources

References:

- George F. Simmons. Calculus with Analytic Geometry (2nd edition) sections 10.6, 10.7.
- Hughes-Hallett. Calculus (6th edition) sections 7.4, 7.2.